Math 514, F13 **Name:** Answer.

Quiz 4 **SSN:** xxx-xx-

Instructions: You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Rolling a fair die 500 times. What is the probability that the number 4 appears more than 100 times?

Let the random variable $X_i = \begin{cases} 1 & \text{if the number 4 appears in i-th rolling} \\ 0 & \text{otherwise} \end{cases}$

clearly $E[X_i] = 1 \cdot \frac{1}{6} + 0 \cdot \frac{5}{6} = \frac{1}{6}$, $Var(X_i) = E[X_i^2] - E[X_i]^2$

$= \frac{1}{6} - (\frac{1}{6})^2 = \frac{5}{36}$

Let $X = \sum_{i=1}^{500} X_i$, then $E[X] = \sum_{i=1}^{500} E[X_i] = 500 \cdot \frac{1}{6} = \frac{250}{3}$

$Var(X) = \sum_{i=1}^{500} Var(X_i) = 500 \cdot \frac{5}{36} = \frac{625}{9}$

Thus, by CLT, we have

$P\{X > 160\} \approx P\{Z > \frac{100 - \frac{250}{3}}{\sqrt{\frac{625}{9}}}\} = P\{Z > \frac{\cancel{50}}{\cancel{3}} \cdot \frac{\cancel{3}}{\cancel{25}}\}$

$= P\{Z > 2\} = 1 - P\{Z \le 2\}$

$= 1 - \Phi(2) \approx 1 - 0.9772 = 0.0228$

Problem 2. (5 points) Let $X(t)$, $t \ge 0$ be a Brownian motion process with drift parameter $\mu = 2$ and variance parameter $\sigma^2 = 16$, and $X(0) = 8$. Find

(a) $E[X(3)]$ and $Var(X(3))$.

$E[X(3)] = E[X(3) - X(0)] + E[X(0)]$

$= 2 \cdot 3 + 8 = 14$

$Var(X(3)) = Var(X(3) - X(0)) + Var(X(0))$

$= 16 \cdot 3 + 0 = 48.$

(b) $P(X(3) > 10)$.

$P(X(3) > 10) = P\{Z > \frac{10-14}{\sqrt{48}}\}$

$\approx P\{Z > -0.5773\}$

$= P\{Z < 0.5773\}$

$= \Phi(0.5773) \approx 0.718$